Poker Theory & Bot Design: Meeting 2

Hand Odds, Pot Odds, Expected Value & Raising

Poker Theory and Desin on Poker Bots Reading Group

February 16, 2026

Lecture 2 Overview

1. Hand Odds: Outs and Equity
2. Rules of 4 & 2
3. Pot Odds & Expected Value
4. What about Raise?
5. Monte Carlo: Equity Simulation
6. Next Time: Kelly Criterion & Bankroll Management

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What Are Hand Odds?

Hand Odds = probability of specific outcomes:

- Drawing a flush after the flop

Given cards I know, what are my chances of winning?

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- Drawing a flush after the flop
- Hitting a straight
- Improving your hand on turn or river

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Outs: Counting Winning Cards

Definition: An *out* is a card improving your hand to (likely) win.

Flush Draw:

- Your hand: K ♥ 10 ♥

Straight Draw:

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- Remaining hearts: $13 - 4 = 9$ outs

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Straight Draw:

- Your hand: 9 ♦ 8 ♦
- Flop: 6 ♠ 7 ♥ 2 ♣
- Need: 5 or 10 $\Rightarrow$ 8 outs (4 fives + 4 tens)

Dirty and Reverse Odds

Definition: A *dirty odd* is some odd that while improving your hand, probably improves the hand of someone else more.

- Your hand: 6 ♣ 7 ♦

Definition: A *reverse (implied) odd* is some odd that, based on behavior of players, probably is beneficial to someone else.

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- Your hand: 7 ♥ 8 ♥
- Flop: 2 ♥ 5 ♣ K ♥
- We would love another heart but probably would the person next to you (who just called), as they may make a higher flush
- So we can't really say that there are 9 outs. How many can we say is a question worth asking?...

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3. Gutshot = 4 outs

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4. Combo draws: subtract overlap

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The Rule of 4 (Flop to River)

Approximation:

$$\text{Equity on flop} \approx \text{Outs} \times 4\%$$

Exact formula (47 cards remain):

$$P(\text{hit} \geq 1 \text{ out in 2 cards}) = 1 - \frac{(47 - O)(46 - O)}{47 \cdot 46}$$

Deriving the Rule of 4

On flop: 5 cards seen (2 hand + 3 board), 47 unknown.

Probability of missing both draws:

$$P(\text{miss both}) = \frac{47 - O}{47} \times \frac{46 - O}{46}$$

Probability of hitting at least one:

$$\begin{aligned} P(\text{hit}) &= 1 - \frac{(47 - O)(46 - O)}{47 \times 46} \\ &\approx 1 - \left(1 - \frac{O}{47}\right)^2 \\ &\approx \frac{2O}{47} \approx \frac{2O}{50} = O \times 4\% \end{aligned}$$

Rule of 4: Error Analysis

Exact vs. Approximate:

Outs	Exact (%)	Rule of 4 (%)	Error (%)
2	8.41	8	-0.42
3	12.48	12	-0.49
4	16.46	16	-0.47
5	20.35	20	-0.36
6	24.14	24	-0.15
7	27.84	28	0.16
8	31.45	32	0.55
9	34.96	36	1.03
10	38.39	40	1.6
11	41.72	44	2.28
12	44.95	48	3.04
13	48.1	52	3.89
14	51.15	56	4.85
15	54.11	60	5.89

The Rule of 2 (Turn to River)

Approximation:

$$\text{Equity on turn} \approx \text{Outs} \times 2\%$$

Exact formula (46 cards remain):

$$P(\text{hit 1 out}) = \frac{O}{46} \approx \frac{O}{50}$$

Rule of 2: Error Analysis

Exact vs. Approximate:

Outs	Exact (%)	Rule of 2 (%)	Error (%)
2	4.34	4	-0.34
3	6.52	6	-0.52
4	8.69	8	-0.69
5	10.86	10	-0.86
6	13.04	12	-1.04
7	15.21	14	-1.22
8	17.39	16	-1.4
9	19.56	18	-1.56
10	21.73	20	-1.74
11	23.91	22	-1.91
12	26.08	24	-2.08
13	28.26	26	-2.27
14	30.43	28	-2.43
15	32.6	30	-2.61

What's Missing?: 2-Card Outs

Rules of 4 and 2 miss: Hands that improve only when <u>both</u> remaining cards help.

Example:

- Your hand: 9 ♦ 8 ♦ (weak)

These 2-card outs contribute $< 1\%$ equity. Given the variables in an actual physical game, one can ignore them safely. But, when it is computers vs computers; these things do matters.

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- You have no outs *yet*
- But if *both* turn *and* river are, say A and K, you make a straight (A-K-9-8-2)

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Pot Odds: Break-Even Threshold

Definition: Minimum win rate needed to break even on a call.

$$\text{Pot Odds} = \frac{\text{Cost to Call}}{\text{Pot Size After Call}}$$

Example:

- Pot: \$100; Opponent bets: \$10

A Simple Decision Rule:

$$\boxed{\text{CALL if Equity} \geq \text{Pot Odds}}$$

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- Pot after call: \$120
- Pot odds: $\frac{10}{120} = 8.33\%$

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Expected Value (EV)

For a call:

$$EV = P(\text{win}) \times (\text{Pot After}) - P(\text{lose}) \times (\text{Your Call})$$

where E = your equity, W = amount you win, L = your call amount.

Decision:

- $EV > 0 \Rightarrow$ CALL

If we make positive EV decisions, we are guaranteed to make money in the long term.

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- $EV > 0 \Rightarrow$ CALL
- $EV < 0 \Rightarrow$ FOLD

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EV Example

Scenario:

- Pot: \$100; Opponent bets: \$20

$$EV = 0.36 \times (100 + 20 + 20) - 0.64 \times 20 = 0.36 \times 140 - 12.8 = 50.4 - 12.8 = +\$37.6$$

Decision: CALL.

We will get the same decision from Pot odds as $\frac{20}{140} * 100 < 36$

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- Equity (by Rule of 4): $9 \times 4\% = 36\%$

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The Three Actions: Fold, Call, Raise

EV for each action:

$$
\begin{aligned}
EV_{\text{fold}} &= 0 \\
EV_{\text{call}} &= E \times W - (1 - E) \times L \\
EV_{\text{raise}} &= ?
\end{aligned}
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Key insight: Raising is a *bet about opponent behavior.*

- If opponent folds: you win the pot immediately

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Key insight: Raising is a *bet about opponent behavior*.

- If opponent folds: you win the pot immediately
- If opponent calls: you compete for larger pot

Opponent's Perspective: Their Pot Odds

Poker is 0 sum $\implies$ your probability of winning is your opponents probability of losing.

If opponent has equity E_{opp}:

$$E_{\text{opp}} = 1 - E_{\text{you}}$$

Opponent's decision when facing your raise:

- Calculate *their* pot odds vs. *their* equity

Strategy: Raise large enough to make opponent's call negative EV or fold.

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- If their equity $<$ their pot odds: they fold (it's bad for them)
- If their equity $\geq$ their pot odds: they call

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Optimal Raise Size

Scenario:

- Current pot: P

If I raise x, I would like:

$$\frac{x}{P+x} < W$$

$$\frac{x}{P+2x} > 1 - W$$

Moving things around,

$$\frac{P}{\frac{1}{1-W} - 2} < x < \frac{P}{\frac{1}{W} - 1}$$

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Moving things around,

$$\frac{P}{\frac{1}{1-W} - 2} < x < \frac{P}{\frac{1}{W} - 1}$$

We get:

$$\text{Raise } \frac{P}{\frac{1}{1-W} - 2} \approx P \cdot \frac{100 - O}{2O - 100} \text{ if} W \geq 0.618 = \frac{1}{\phi}$$

Note, this only works if your opponent plays rationally and doesn't bluff as well as if we are not in one of the edge cases.

Raise EV Analysis

When opponent calls your raise:

$$EV_{\text{raise}} = P(\text{opp calls}) \times [E \times (\text{pot} + 2 \text{ raises}) - (1 - E) \times \text{raise}]$$

When opponent folds:

$$EV_{\text{raise}} = P(\text{opp folds}) \times \text{current pot}$$

Notice: We are yet to talk about the probability of opponent folding and calling. We will talk about them next time.

Tournament Context: n-Way Pots

In cash games: Opponent's equity $= 1 -$ yours.

In tournaments with n players still deciding:

$$E_{\text{opp}} \approx \frac{1 - E_{\text{you}}}{n - 1}$$

(Simplified: remaining equity split equally among other players.)

Implication:

- In 4-way: opponent's equity $= \frac{1-0.6}{3} = 13.3\%$ (if you're 60%)

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- In 4-way: opponent's equity $= \frac{1-0.6}{3} = 13.3\%$ (if you're 60%)
- Much weaker threshold for them to call your raise

Tournament Context: n-Way Pots

In cash games: Opponent's equity $= 1 -$ yours.

In tournaments with n players still deciding:

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- Is it more profitable to raise in multiway pots?
- We will see that this idea doesn't really work...

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1. Hand Odds: Outs and Equity
2. Rules of 4 & 2
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4. What about Raise?
5. Monte Carlo: Equity Simulation
6. Next Time: Kelly Criterion & Bankroll Management

Monte Carlo Overview

Why simulate?

- Unusual outs or blocking scenarios

Algorithm:

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4. Repeat $N \geq 10000$ times
5. Equity $= \frac{\text{wins}}{N}$

Math Bot Strategy

Decision Rules:

1. Estimate equity

Properties:

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5. **FOLD** otherwise

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Decision Rules:

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Properties:

- Boring but mathematically sound
- Beats weak opponents

Math Bot Implementation I

```python
import random
from pypokerengine.players import BasePokerPlayer
from pypokerengine.utils.card_utils import gen_cards, estimate_hole_card_win_rate
from pypokerengine.utils.card_utils import _pick_unused_card
from pypokerengine.engine.hand_evaluator import HandEvaluator


class MathBot(BasePokerPlayer):

    SIMULATIONS = 10000
    RAISE_THRESHOLD = 0.61

    def estimate_equity(self, hole_card, community_card, nb_players):
        """
        Monte Carlo equity estimation
        """
        win_rate = estimate_hole_card_win_rate(
            nb_simulation=self.SIMULATIONS,
            nb_player=nb_players,
            hole_card=gen_cards(hole_card),
```

Math Bot Implementation II

```
            community_card=gen_cards(community_card)
        )
        return win_rate

    def calculate_pot_odds(self, call_amount, pot):
        if call_amount == 0:
            return 0
        return call_amount / float(pot + call_amount)

    def declare_action(self, valid_actions, hole_card, round_state):

        community_card = round_state['community_card']
        pot = round_state['pot']['main']['amount']
        nb_players = len(round_state['seats'])

        # Extract call info
        call_action = next(a for a in valid_actions if a['action'] == 'call')
        call_amount = call_action['amount']

        raise_action = next(a for a in valid_actions if a['action'] == 'raise')
```

Math Bot Implementation III

```
        # Estimate equity
        equity = self.estimate_equity(hole_card, community_card, nb_players)

        # Compute pot odds
        pot_odds = self.calculate_pot_odds(call_amount, pot)

        # Decision logic
        if equity >= pot_odds:

            # Raise if strong enough
            if equity >= self.RAISE_THRESHOLD and raise_action['amount']['min'] !=
                -1:
                min_raise = raise_action['amount']['min']
                max_raise = raise_action['amount']['max']

                # Simple proportional raise sizing
                raise_amount = int(pot * equity)

                # Clamp to legal range
```

Math Bot Implementation IV

```
                raise_amount = max(min_raise, min(raise_amount, max_raise))

                return "raise", raise_amount

            return "call", call_amount

        return "fold", 0


    # Required abstract methods
    def receive_game_start_message(self, game_info):
        pass

    def receive_round_start_message(self, round_count, hole_card, seats):
        pass

    def receive_street_start_message(self, street, round_state):
        pass

    def receive_game_update_message(self, action, round_state):
```

Math Bot Implementation V

```
        pass

    def receive_round_result_message(self, winners, hand_info, round_state):
        pass
```

Sadly...

While we do beat the "Angry Uncle" bot more often than not, we also do lose with a significant probability.
We would like to

1. Understand why that is the case

Sadly...

While we do beat the "Angry Uncle" bot more often than not, we also do lose with a significant probability.
We would like to

1. Understand why that is the case
2. And fix it!

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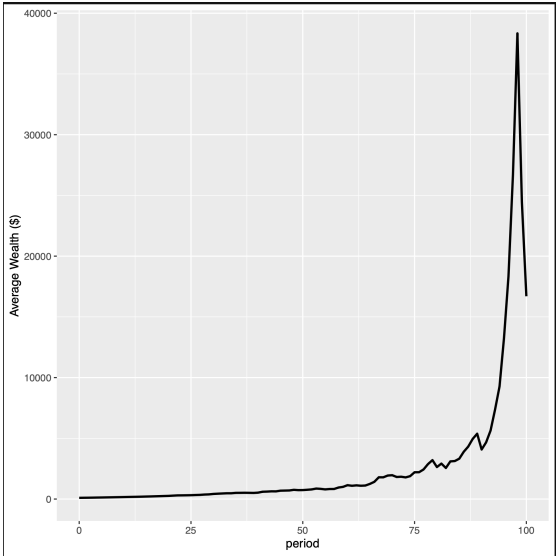
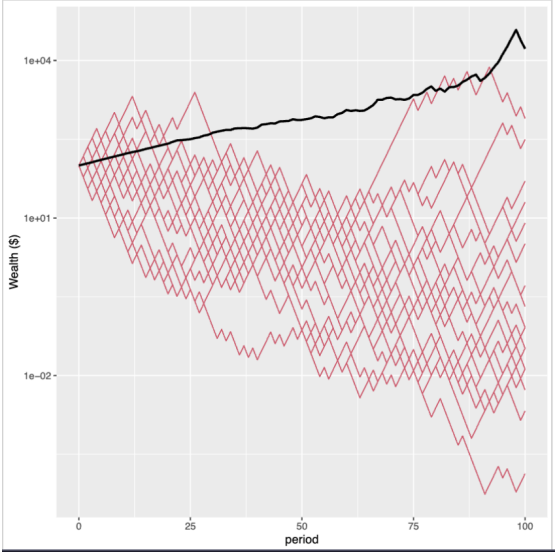
Dealing With Uncertainity

Suppose you have \$100 and are offered a gamble involving a series of coin flips. For each flip, heads will increase your wealth by 50%. Tails will decrease it by 40%. Flip 100 times.
What will happen? For that first flip, you have a 50% chance of a \$50 gain, and a 50% chance of a \$40 loss. Your expected gain is \$5 or 5% of your wealth. The absolute size of the stake for future flips will depend on past flips, but for every flip you have the same expected gain of 5% of your wealth.
Should you take the bet?

Some Simulations!





Losing While Winning

This forces us to consider the counter intuitive idea that while our EV might be positive, we can still lose wealth!

This extreme diversion of mean and median is called *non-ergodicity*.

One way to justify the failure of EV is that EV expects infinite replays from the same state as well as infinite resource. Neither are very practical in above scenario (or in poker).

Furthermore, this opens us up to downside where we go all in on a positive EV bet and the cards don't turn up our way.

Kelly Criterion: Optimal Bet Sizing

Problem: Given probability of winning, How much of bankroll should you risk?

This exact problem was solved by **John Larry Kelly Jr**. After all, who knows more about risk than a man who smoked 6 packs of cigarettes a day and died of stroke at 41, while walking on a Manhattan street.
We will come back to this in 2 week trying to re-derive Kelly's formula and talk about Bayesian predictions on the opponents hands.

Questions?

Next Week: Balanced ranges, Solving the Pre-Flop and game theory.